

Frozen BERT adds 11.6 accuracy points over local GloVe context

67.1% vs 55.5%; paired 95% interval for GloVe minus BERT: [-16.9, -6.6] points

Hypothesis

A single static vector conflates a word’s meanings; WiC asks whether sentence context separates them (Camacho-Collados & Pilehvar, 2018; Pilehvar & Camacho-Collados, 2019).

Prediction: frozen BERT target similarity will separate WiC labels better than either a context-free GloVe target or an average of nearby GloVe words.

Methodology

Data. Official SuperGLUE v2 WiC: 5,428 balanced training pairs and one final evaluation on 638 balanced validation pairs. The 1,400-row test split has no public labels and is not reported.

Static representations. GloVe 6B 300d (Pennington et al., 2014): (i) the lemma vector on both sides; (ii) mean in-vocabulary context vectors after excluding the marked target. Candidate windows were ± 2 , ± 5 , ± 10 , and sentence.

Contextual representation. Unfine-tuned BERT-base-uncased (Devlin et al., 2019). Fast tokenizer offsets select every WordPiece overlapping the supplied half-open character span; WordPieces are mean pooled. Candidates: final layer and mean of the last four layers.

Decision rule. Cosine similarity $\geq$ a global threshold means same-sense. Thresholds maximize macro F1 on training data only.

Selection stayed inside training

5,428 train

↓ stratified, seed 2026

4,342 tune threshold → 1,086 rank candidates

↓ freeze ± 2 and mean-last-four

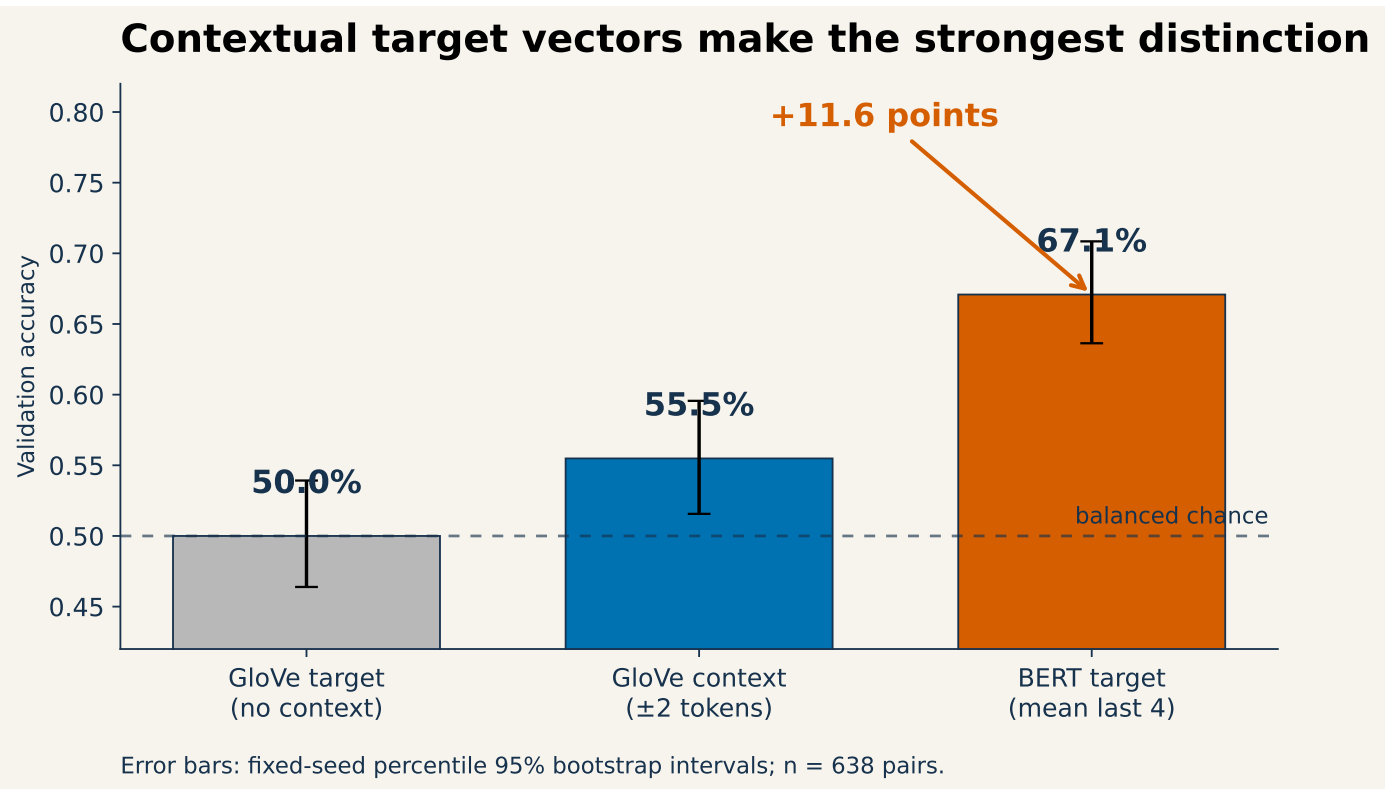
refit thresholds on all train

↓ once

638 validation

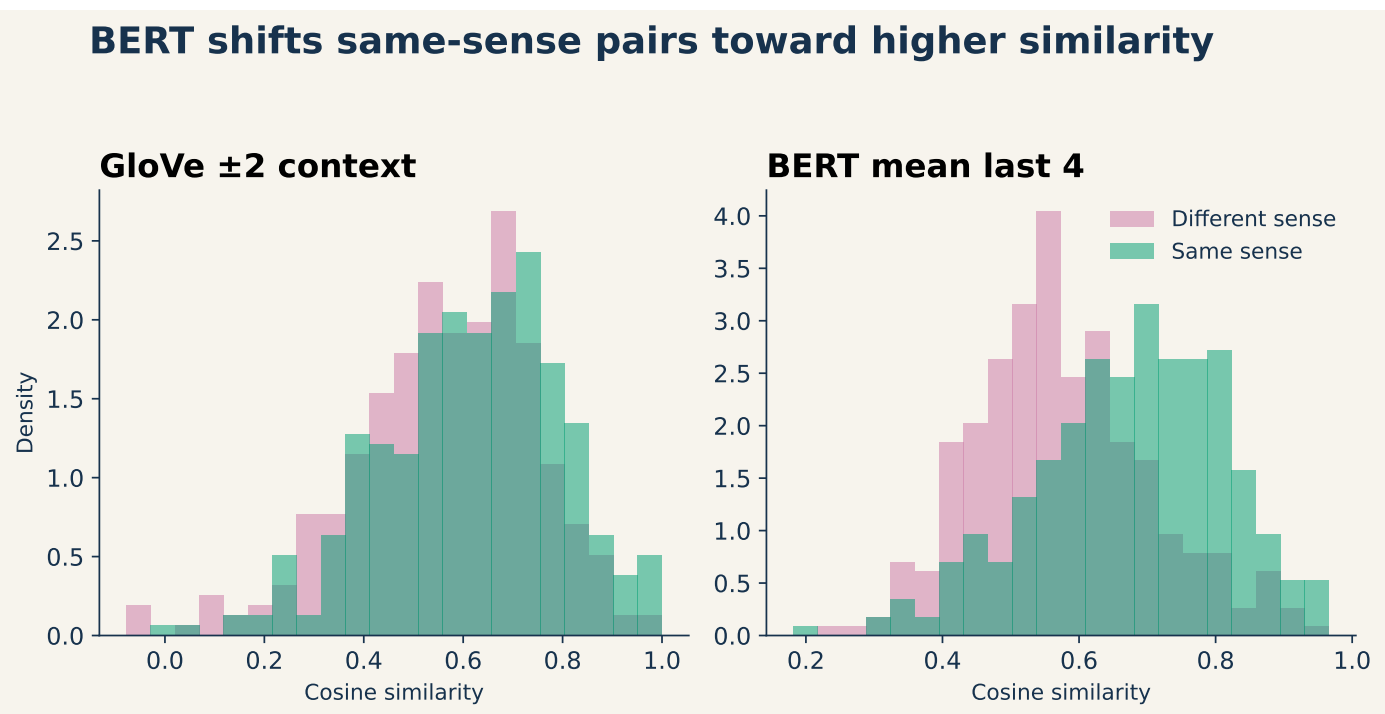
The machine-readable ledger is hashed before validation scoring.

Results: context changes the geometry



BERT reaches 67.1% accuracy and 66.7% macro F1. The target-only diagnostic is exactly constant; nearby static context restores limited signal.

Same-sense scores separate more clearly



GloVe’s label distributions overlap heavily. BERT shifts same-sense pairs toward higher similarity, while substantial overlap remains (ROC-AUC 0.716).

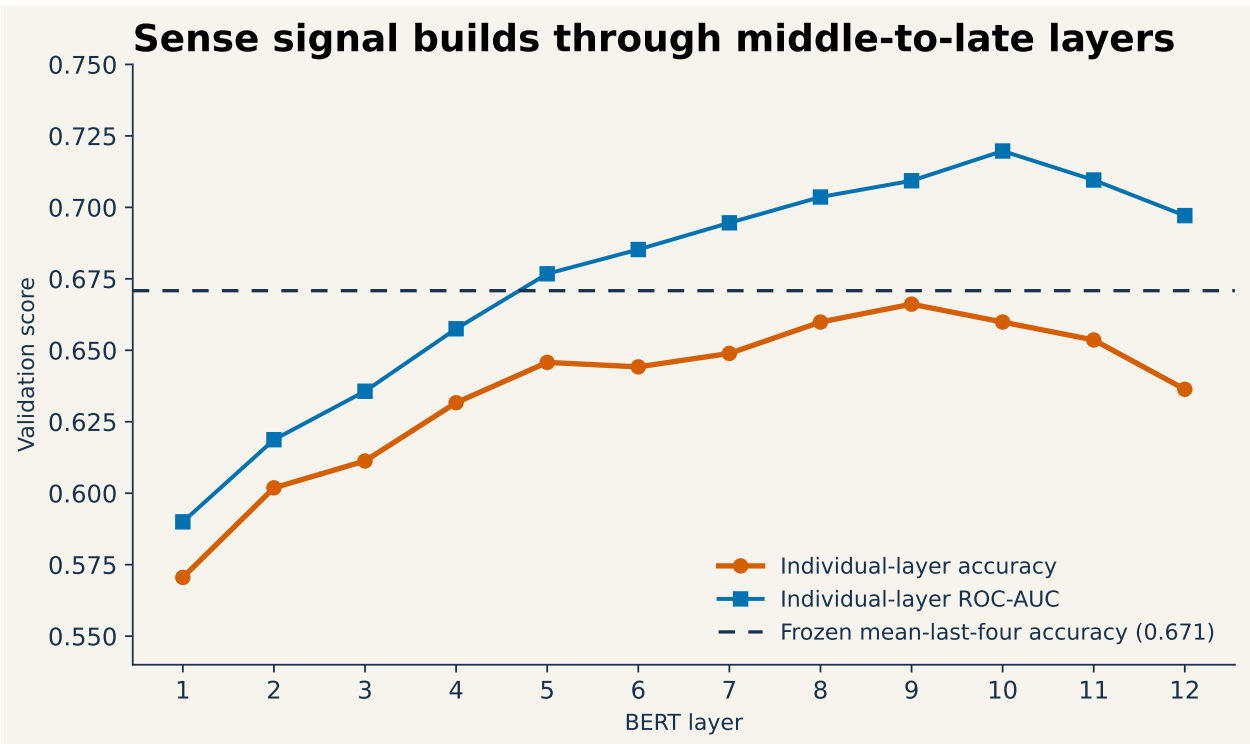
Paired evidence, not decorative error bars

The BERT-minus-context gain is 74 additional correct pairs. In 1,000 paired bootstrap resamples, the reverse contrast (GloVe minus BERT) remains below zero: accuracy [-16.9, -6.6] points and macro F1 [-16.9, -6.6] points.

The systems fail in opposite directions

BERT recovers 77.4% of same-sense pairs but 56.7% of different-sense pairs. GloVe context recovers 46.4% and 64.6%, respectively. These class skews are thresholded observations, not intrinsic representation preferences.

Where the signal appears



Individual-layer accuracy rises broadly through layer 9 and falls at layer 12. This is secondary validation analysis; it did not redefine the frozen primary representation.

One success, one warning

Context helps—until closely related usages cross the decision boundary

BERT sees the shared sense
1 Rust remover.
2 Paint remover.

Gold: same
BERT: same (0.96)
GloVe: different (0.35)

A confident miss
1 The ball at the base of the thumb.
2 He stood on the balls of his feet.

Gold: same
BERT: different (0.18)
GloVe: same (0.83)

BERT is uniquely correct on 178 pairs; GloVe context is uniquely correct on 104. A contextual representation improves separation, but it does not settle fine-grained sense boundaries.

Limitations

- One balanced English benchmark and 638 validation pairs.
- Frozen BERT features, not fine-tuning or a modern encoder comparison.
- GloVe context is an unweighted lexical mean; syntax is absent.
- Thresholded cosine tests geometry, not every fact encoded in a vector.
- 210 BERT errors remain; selected cases suggest hypotheses, not causal explanations.

Conclusion

Contextualization matters here. Under one locked protocol, BERT target vectors discriminate word meaning substantially better than local static context. The middle-to-late layer pattern and confident errors also show that “more contextual” is not the same as “ambiguity solved.”